



❖ 1.2.5: COMMON CYBER ATTACKS

Cybercriminals commit a variety of cybercrimes to access sensitive data, disrupt computer operations, gain unauthorised access and/or use or damage any digital resource. These attacks exploit **vulnerabilities** in security systems or use social engineering to trick end-users into divulging sensitive information and they are aimed at individuals, businesses, schools and government agencies.

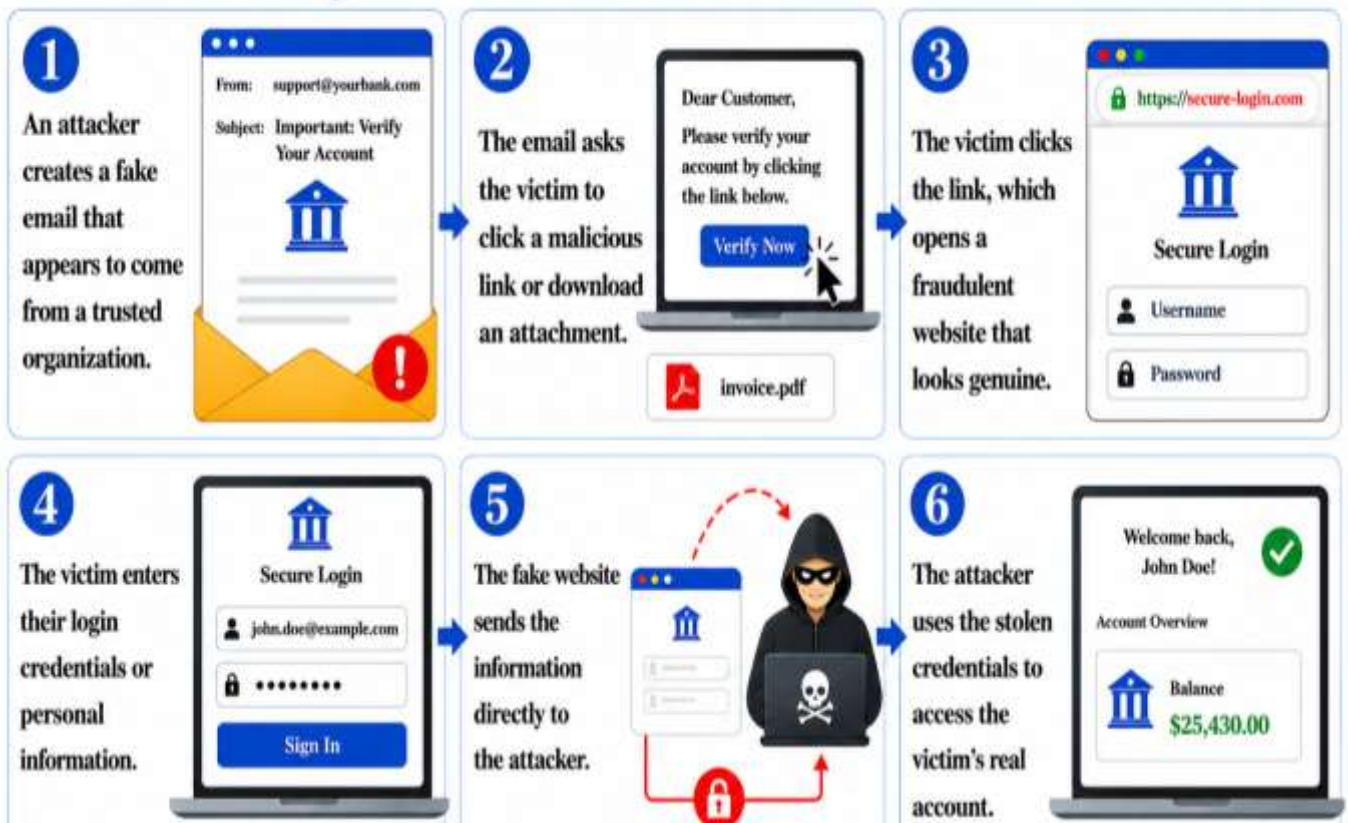
■ Phishing Attacks

A **Phishing** attack is a cyber-attack in which attackers use spoofed emails, forged websites or deceptive messages to ask people for usernames, passwords and banking information. In this attack, the attacker impersonates a trusted **organization**; may be from a bank, online shopping site or any social media platform. **Social engineering** is the most prevalent type of social engineering — users are more easily lured into clicking on something if they think they can trust it than exploiting an actual technical weakness.

► **Example: How a Phishing Attack Happens:** An attacker creates a fake email that appears to come from a trusted organization. The email asks the victim to click a malicious link or download an attachment. The victim clicks the link, which opens a fraudulent website that looks genuine. The victim enters their login credentials or personal information. The fake website sends the information directly to the attacker. The attacker uses the stolen credentials to access the victim's real account.



Example: How a Phishing Attack Happens



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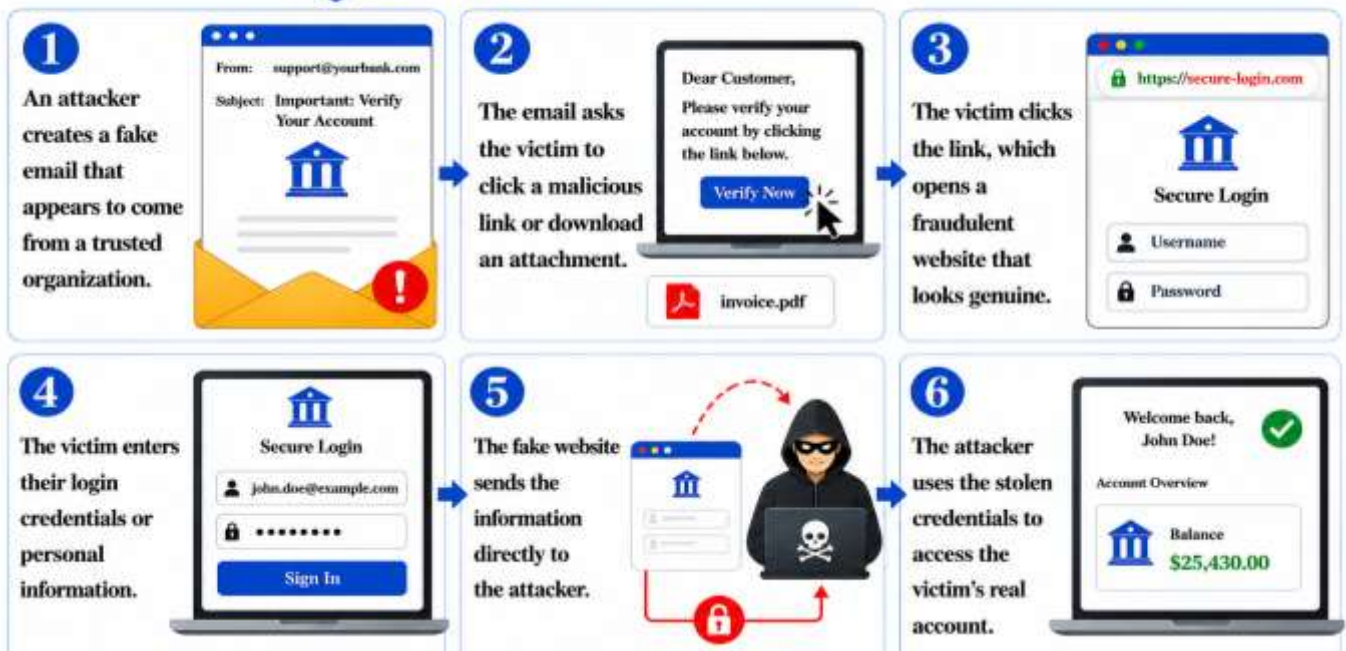
■ DoS and DDoS Attacks

Denial of Service (DoS) attack is an attempt to make computer, server or network resource unavailable. By sending superfluous **requests** with the goal victim, machine won't be able to process them leading to unavailability. A **Distributed Denial of Service (DDoS)** attack operates on the same principle but utilizes thousands of compromised devices from a range of geographic locations, making it an altogether bigger and much more difficult to eradicate attack. These are a new form of **attack** that disrupts online services so legitimate users cannot access the service.

► **Example: How a DDoS Attack Happens:** An attacker controls thousands of compromised computers and IoT devices. These devices simultaneously send millions of fake requests to a website. The web server tries to process every request. The server becomes overloaded and its resources are exhausted. Genuine users cannot access the website because the server is too busy responding to fake traffic. The service remains unavailable until the malicious traffic is filtered or blocked.



Example: How a Phishing Attack Happens



■ Man-in-the-Middle (MitM)

A **Man-in-the-Middle (MitM)** attack is a type of cyberattack where an attacker secretly places themselves between two communicating parties (such as a user and a website) to intercept, read, modify, or steal information without the users knowing.

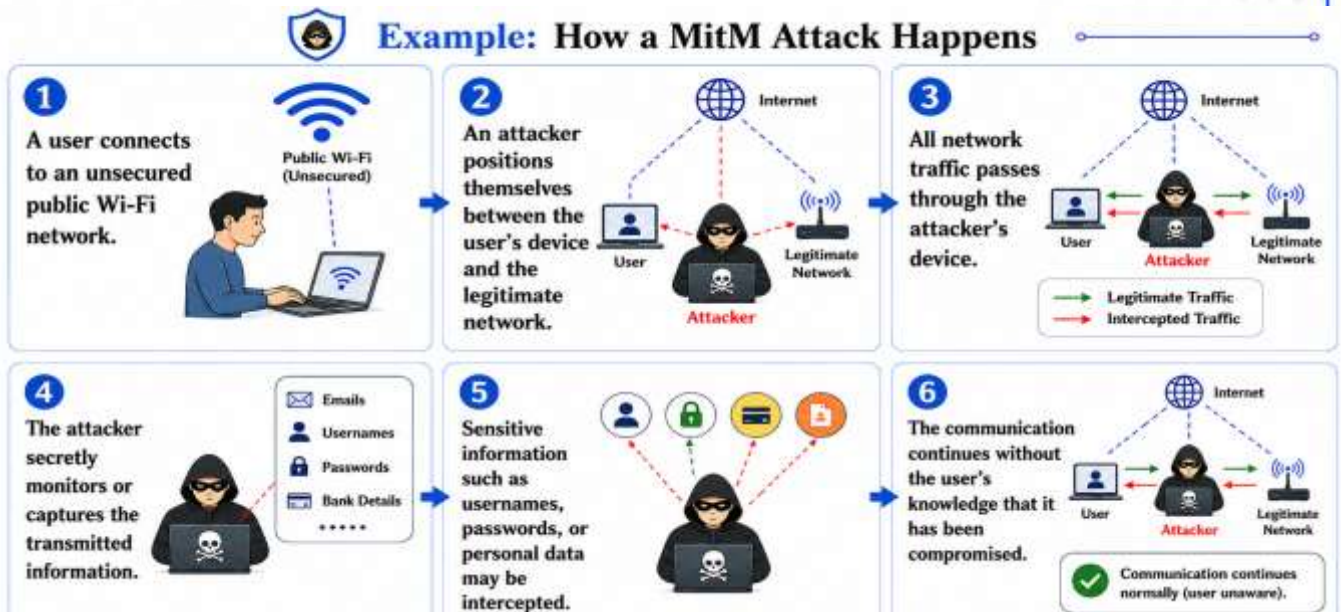
► **Example: How a MitM Attack Happens:** A user connects to an unsecured public Wi-Fi network. An attacker positions themselves between the user's device and the legitimate network. All network traffic passes through the attacker's device. The attacker secretly monitors or captures the transmitted

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information. Sensitive information such as usernames, passwords, or personal data may be intercepted. The communication continues without the user's knowledge that it has been compromised.



■ ▲ SQL Injection (SQLi)

SQL Injection (SQLi) is a type of cyberattack where an attacker inserts malicious SQL commands into an application's input fields to access, modify, or delete information stored in a **database**.

► **Example: How a SQL Injection Attack Happens:** A user opens a website's login page or search form. An attacker enters malicious input into an input field instead of normal information. The website fails to validate user input before sending it to the database. The database incorrectly interprets the malicious input as part of an SQL query. The database returns information or performs an action that the attacker is not authorized to access. As a result, sensitive data may be exposed or unauthorized access may be granted.

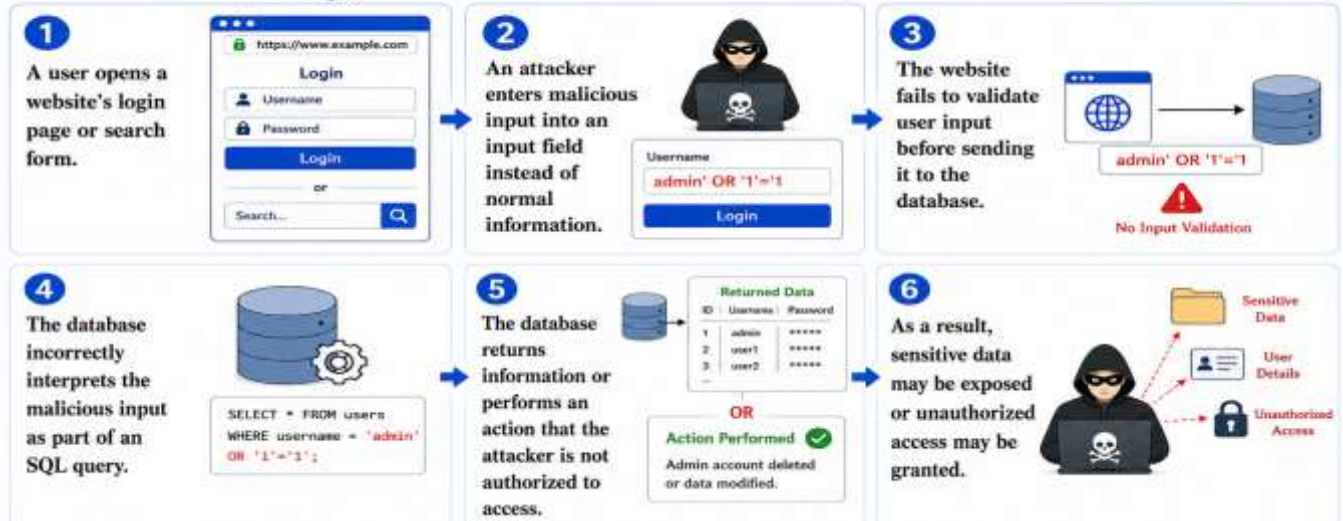
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Example: How a SQL Injection Attack Happens



❖ 1.2.6: ADVANCED CYBER ATTACKS

Cybercriminals, security controls, unauthorized access: Cybercriminals are continually innovating more complex forms of attack in order to evade security controls, steal sensitive information and achieve unauthorized access to computer systems. Unlike traditional **cyberattacks**, these are characterized by collecting relevant data on the target using high-tech methods over many months before launching a full-scale attack or exploiting unknown software vulnerabilities. Knowledge of these types of **cyberattacks** is crucial for organizations and individuals to improve their own cybersecurity program and to respond adequately to emerging threats.

► **Example:** A multinational company discovers that confidential product designs have been secretly copied over several months. The investigation reveals that attackers first infected employee computers with malicious advertisements, then stole weak passwords to access internal systems. They exploited an unknown software vulnerability to install hidden malware, quietly monitored employee communications, and controlled compromised computers through a botnet. Fortunately, the cybersecurity team detected the unusual activity, isolated the infected systems, updated vulnerable software, and prevented further data theft.

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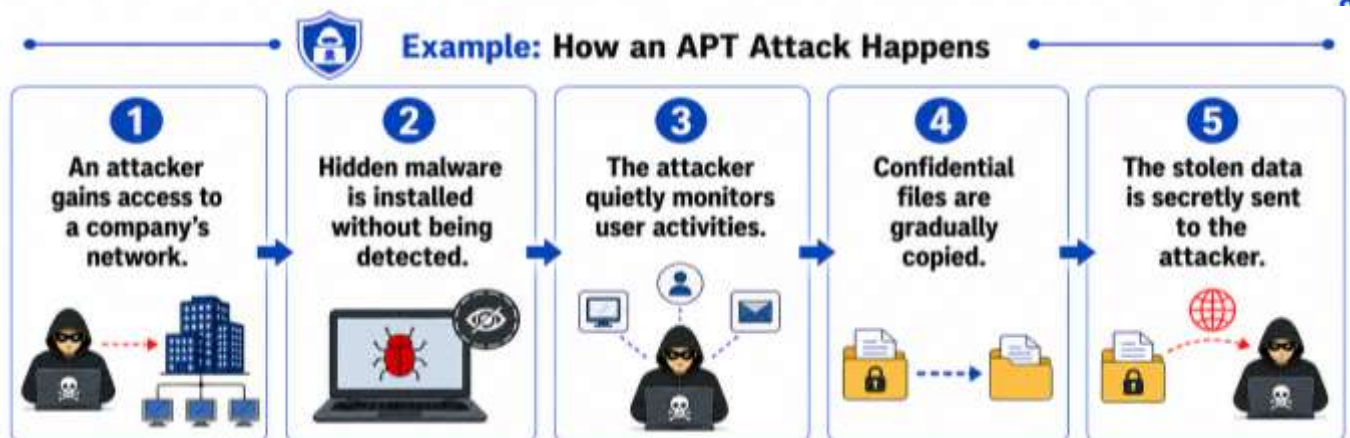
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■ Advanced Persistent Threat (APT)

APT-Advanced Persistent Threat (APT) is a cyberattack in which intruders gain access to a computer network and remain there for a prolonged period of time without being detected, stealing sensitive information along the way.

► **Example: How an APT Attack Happens:** An attacker gains access to a company's network. Hidden malware is installed without being detected. The attacker quietly monitors user activities. Confidential files are gradually copied. The stolen data is secretly sent to the attacker.



■ Zero-day Exploits

A **Zero-Day Exploit** is a type of cyberattack that takes advantage of a previously unknown security vulnerability in software, hardware, or systems before the developer

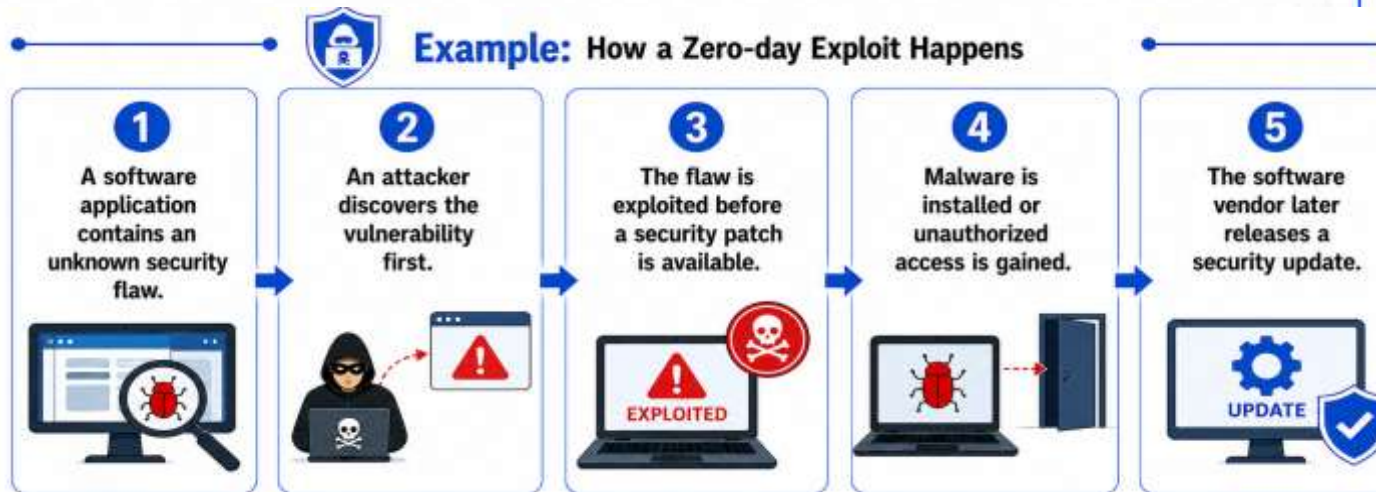
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has released a fix or patch. The term “**zero-day**” means that the developers have had **zero days to address the vulnerability** before attackers discover and exploit it.

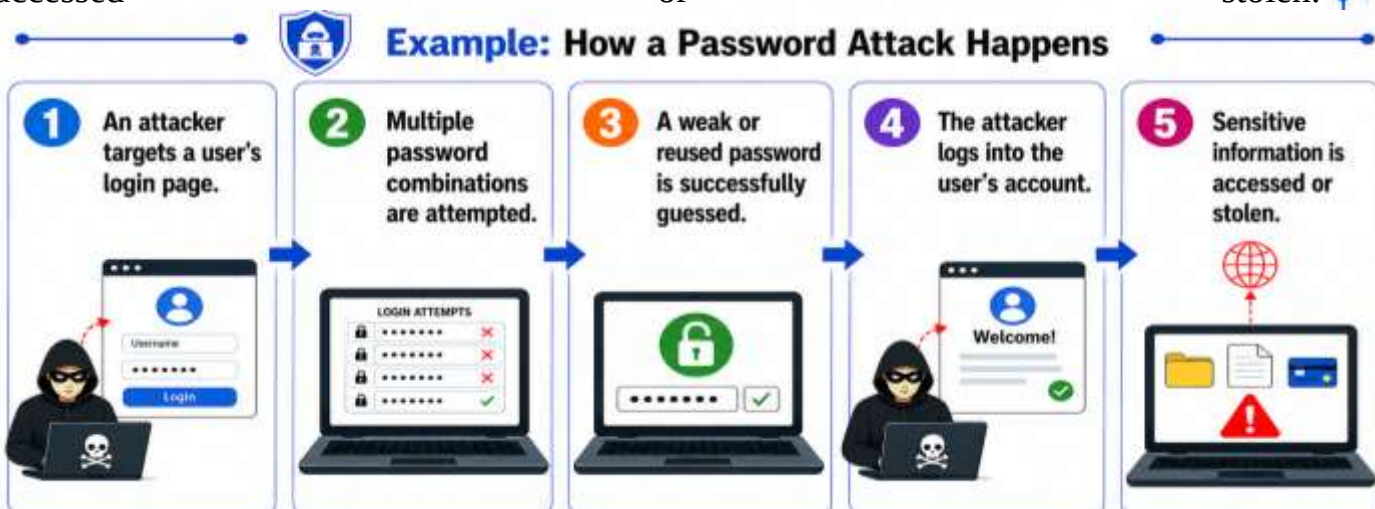
► **Example: How a Zero-day Exploit Happens:** A software application contains an unknown security flaw. An attacker discovers the vulnerability first. The flaw is exploited before a security patch is available. Malware is installed or unauthorized access is gained. The software vendor later releases a security update.



Password Attacks

Password attack, login credentials, authentication: A **Password Attack** seeks to acquire a user's password, login credentials or authentication data so that the attacker can use this in order to gain entry without authorization.

► **Example: How a Password Attack Happens:** An attacker targets a user's login page. Multiple password combinations are attempted. A weak or reused password is successfully guessed. The attacker logs into the user's account. Sensitive information is accessed or stolen.



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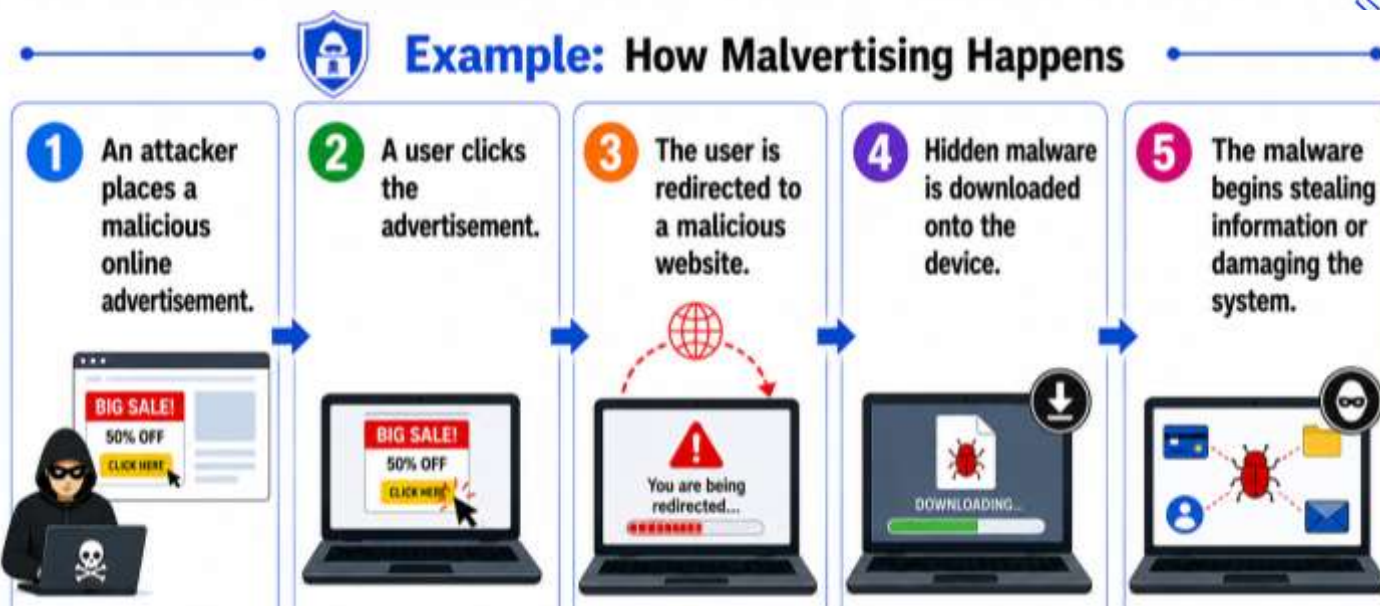


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■ Malvertising

Malvertising (Malicious Advertising) is a type of cyberattack where attackers use online advertisements to spread malware, steal information, or redirect users to harmful websites.

► **Example: How Malvertising Happens:** An attacker places a malicious online advertisement. A user clicks the advertisement. The user is redirected to a malicious website. Hidden malware is downloaded onto the device. The malware begins stealing information or damaging the system.



■ Eavesdropping

Eavesdropping, network traffic, electronic communications: **Eavesdropping** is a type of cyberattack that occurs when an attacker secretly intercepts, monitors or listens to electronic communications.

► **Example: How Eavesdropping Happens:** A user connects to an unsecured public Wi-Fi network. An attacker intercepts the network traffic. Login credentials and sensitive data are captured. The user remains unaware of the interception. The attacker later misuses the

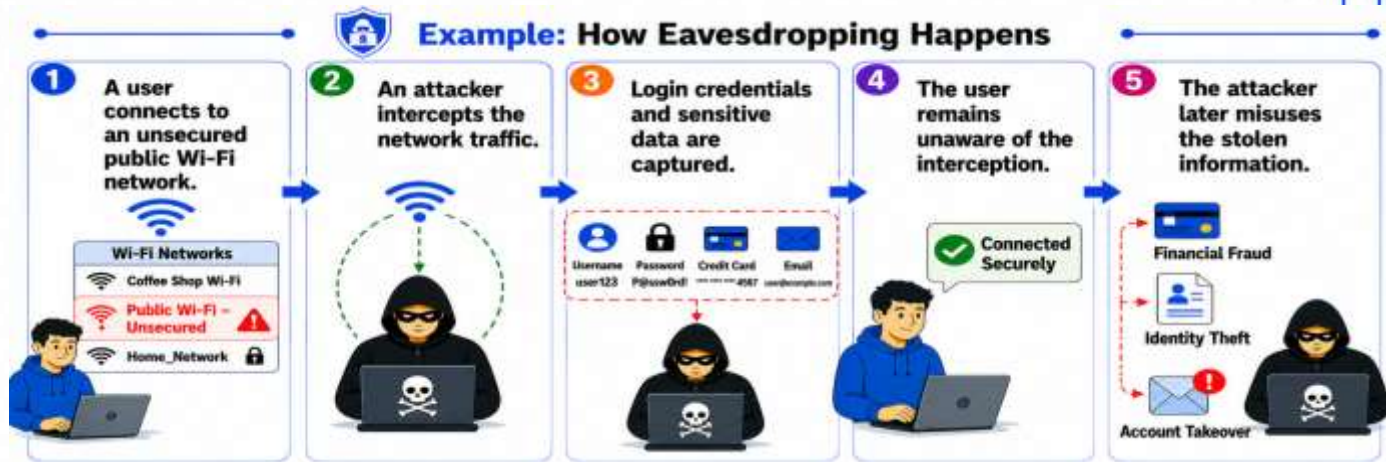
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stolen

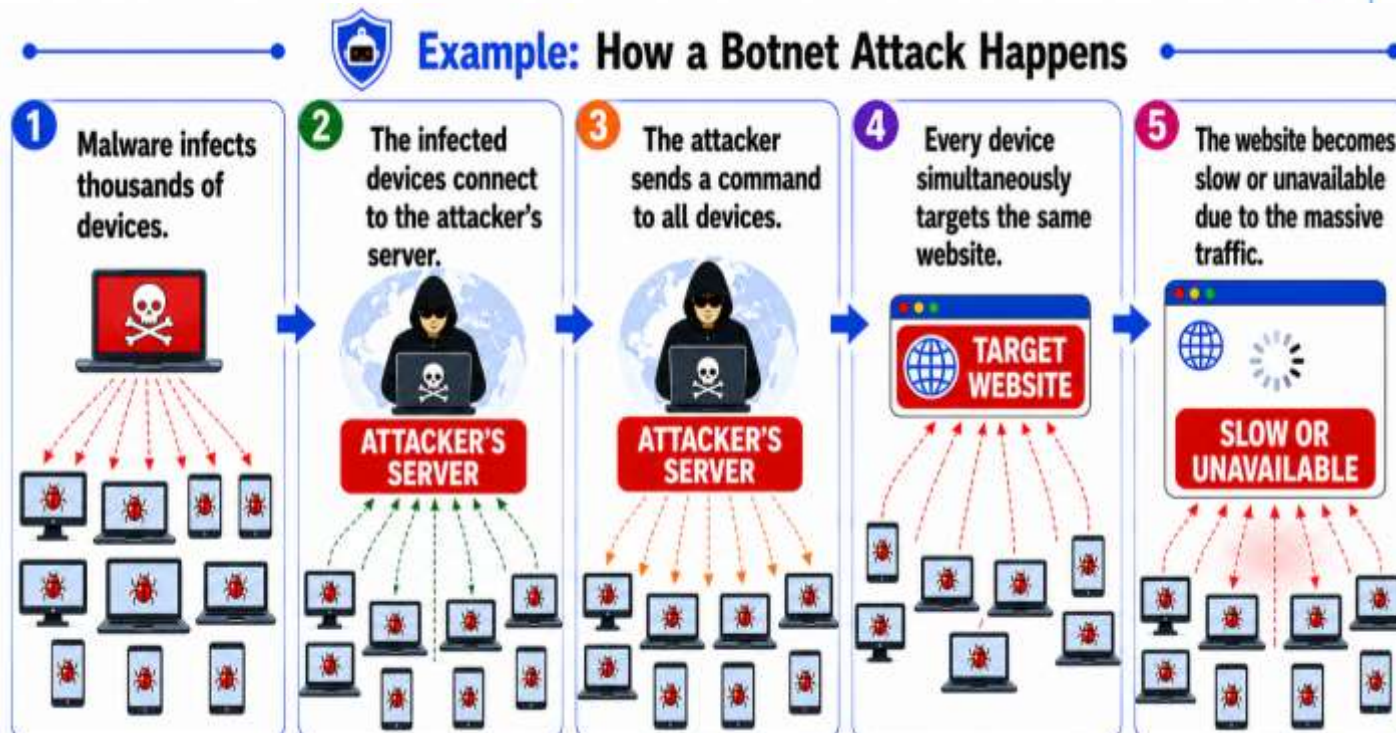
information.



Botnets

Botnet, Internet of Things (IoT), cybercriminals: **Botnet** is a network of infected devices, compromised computers or Internet of Things (IoT) devices that are secretly controlled by cybercriminals.

Example: How a Botnet Attack Happens: Malware infects thousands of devices. The infected devices connect to the attacker's server. The attacker sends a command to all devices. Every device simultaneously targets the same website. The website becomes slow or unavailable due to the massive traffic.



SESSION REVIEW



1.2.5 COMMON CYBER ATTACKS

Phishing Attacks



- Uses spoofed emails, forged websites, or deceptive messages.

DoS and DDoS Attacks



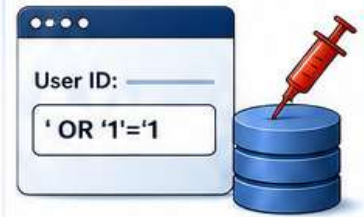
- Attempts to make computers, servers, or networks unavailable.

Man-in-the-Middle (MitM)



- Attacker secretly intercepts communication between two parties.

SQL Injection (SQLi)



- Inserts malicious SQL commands into application input fields.

1.2.6 ADVANCED CYBER ATTACKS

Advanced Persistent Threat (APT)



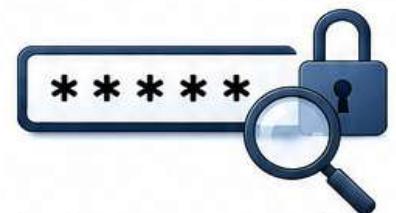
- Intruders remain undetected while stealing sensitive information.

Zero-day Exploits



- Exploits unknown security vulnerabilities before a patch is released.

Password Attacks



- Seeks user passwords, login credentials, or authentication data.

Malvertising



- Uses online advertisements to spread malware or steal information.

Eavesdropping



- Secretly intercepts, monitors, or listens to electronic communications.

Botnets



- Network of infected devices secretly controlled by cybercriminals.